Section 0 – Title and contact Information

- **Title of the proposed solution track:** Scaling Integrated Digital Climate Advisory Services for 6 Million Farmers through PPP in Kenya
- **Title of the AoW2 Flagship that the solution track aligns with:** Flagship 1.1: Scaling Digital Climate Services for Farmers and Enterprises (*primary; with potential contribution to other tracks*)
- **Lead Scientist(s) / Contact Person(s) / Center(s):** Aniruddha Ghosh/Alliance of Bioversity and CIAT
- **Emails of the Lead Scientist(s) / Contact Person(s) / Center(s):** a.ghosh@cgiar.org
- **Key CGIAR Center Partners (in the case of multi-Center EoIs):** ILRI, IFPRI
- **Technical link to other Science Programs – name the corresponding Science Programs by name:** Climate Action (CA), Sustainable Farming Program (SFP), Sustainable Animal and Aquatic Foods (SAAF); Digital Transformation (DTA); and Gender, Equity and Inclusion (GESI).

Section 1 – Innovation Package, Value Proposition & Alignment

- **The problem to be solved:** Kenya already has strong institutions for agriculture and climate information—KALRO for agricultural advisories and the KMD for forecasts—yet these systems remain fragmented. Forecast data flows partially between KMD and KALRO (e.g., seven-day outlooks for selected counties), but translation into timely, value chain–specific advisories is limited, and downstream actors such as extension agents, county programs, and private networks do not consistently benefit. At the same time, millions of smallholder farmers—especially women, youth, and pastoralists in arid counties—struggle to access reliable, localized advice to manage droughts, erratic rains, or heat stress on crops and livestock. Traditional in-person extension reaches fewer than 20% of farmers, leaving the majority without actionable guidance. Kenya's farmer registry of 7 million+ and emerging digital public infrastructure (DPI) for agriculture provide the building blocks: farmer profiling, an integrated central database, and interoperable digital tools for data-driven innovations. Yet without integration and development of agricultural use cases, these assets remain underutilized. The track directly addresses this gap by institutionalizing digital climate services within Kenya's public extension system, creating a sustainable model for nationwide delivery.
- **Your scaling ambition:** By 2028, this track aims to deliver reliable, localized climate advisories to at least six million smallholder farmers in Kenya, representing nearly the entire national farmer registry. The ambition is not only to expand numerical reach but to ensure equitable impact, with at least 50% women and 40% youth among active users. Farmers will receive ≥10 actionable advisories per season, spanning 20 crops, livestock, and rangeland management, enabling them to anticipate climate shocks and adopt timely practices. The track will contribute most directly to S4I KPIs on farmers reached, adoption of CGIAR innovations, and hectares under climate-smart practices—with a target of 2 million hectares managed under improved practices by 2028. Beyond numbers, the ambition is to institutionalize a national digital public infrastructure for agriculture, embedding advisory services into county workplans (NAVCDP, FSRP) and the National

Extension Strategy, thereby ensuring that impact is sustained, government-owned, and scalable to other contexts.

- **Innovation & Value Proposition** – The innovation lies in creating an integrated, end-to-end advisory system that connects Kenya's existing assets—KMD forecasts, KALRO's extension platform, and the national farmer registry—into a a unified digital public infrastructure for agriculture, connecting forecasts to advisories and last-mile delivery. Unlike stand-alone tools, this bundle upgrades fragmented services into a coherent DPI for agriculture, where forecasts flow through automated APIs, are translated into crop and livestock advisories, and reach farmers via SMS, USSD, IVR, and agent networks. Novel elements include AI-assisted translation and personalization of advisories into multiple local languages, with human validation to ensure accuracy and trust, and data-driven farmer segmentation that tailors messages to production systems and value chains. For institutional partners, the system enhances capacity to deliver services at scale—reducing extension costs from US$20–35 per farmer to less than US$1—while generating real-time data for planning. For farmers, it improves productivity, resilience, and decision-making, especially for women, youth, and pastoralists. The track draws on CGIAR science programs—CA (better forecast), SFP (agronomy), SAAF(livestock), DTA (generative AI and HCD), and GESI (gender-responsive and inclusivity).
- **Customers & Alignment** – The main institutional customers are Kenya's Ministry of Agriculture and Livestock Development (MOALD), KALRO, and the KMD, supported by county agriculture departments that deliver extension. Major donor programs—WB's NAVCDP and FSRP, IFAD's PPP on CIS, and FAO and WFP FtMA—will benefit from this solution track in their operations. NGOs and digital service providers such as Digital Green and Kuza are implementation partners. End-users are smallholder farmers (with emphasis on women, youth, and pastoralists), supported by cooperatives, agro-dealers, and SMEs. These customers and users form the scaling pathway that the track strengthens, directly advancing Flagship 1.1 of Grand Challenge 1 by institutionalizing digital climate services, mobilizing donor alignment, and ensuring that millions of farmers can benefit at scale.
- **Evidence of Impact** – A meta-analysis of digital information interventions in SSA found average yield gains of 6–9% and significant increases in fertilizer adoption when advisories were delivered through multiple channels (Beach et al., 2025). In India, Fabregas et al. (2019) observed that scaled digital advisory programs generated ~4% average yield increases, highlighting modest but consistent effects at scale. Evidence from Kenya is equally compelling: KALRO and KMD have jointly delivered SMS agro-weather advisories to over a million farmers, with 50% of surveyed farmers willing to pay ~$0.58 per month (Basis, 2025). Independent evaluations of the Digital Weather Advisory Service show farmers achieved a 12% yield increase at a cost of $0.56 per farmer, with 88% reporting the forecasts as accurate and useful (TomorrowNow, 2024). These findings validate the feasibility of the KAOP/DCAS innovation bundle, which addresses adoption drivers by integrating trusted institutions, local content, multiple delivery channels, and county-level extension agents. Limitations remain in ensuring consistent seasonal use and equitable access among low-literacy and remote populations.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

- **Scaling Pathway & Sustainability** – The track will expand through a hybrid PPP model, anchored in Kenya's national digital agriculture infrastructure. At the core, KALRO's KAOP integrates forecasts from KMD and global models, linked to the national farmer registry of 7 million+ farmers. This provides the backbone for a national rollout, while county governments and private service providers ensure last-mile delivery.

The scaling sequence is threefold:

- National rollout through KAOP, delivering SMS, USSD, and IVR advisories in multiple local languages.
- County reinforcement by embedding climate service teams within NAVCDP/FSRP structures, integrating the government's agripreneur network and local extension agents.
- Private and development linkages, where telecom operators, agri-businesses, and NGOs (e.g. Digital Green, Kuza, AGRA, FtMA) expand reach and bundle advisories with complementary services such as insurance or market access.

Financial sustainability is pursued through blended financing:

- Public funding: core services (weather alerts, basic advisories) financed through MOALD budgets, county allocations under FLLoCA, and WB programs.
- Donor/IFI support: catalytic investments from AICCRA III and IFAD/Gates Foundation's upcoming PPP model in WCIS.
- Private contributions: sponsored SMS campaigns, telco CSR, and freemium models for value-added services (personalized crop calendars, market linkages).

Milestones: By 2027, at least 30% of operational costs will be covered by domestic public or private sources. By 2028, the proposed DCAS will be fully institutionalized via the National Extension Strategy or equivalent, with a roadmap for sustained government financing. With delivery costs projected at <$1 per farmer annually, the service is both viable and scalable.

- **Partnerships & Leverage** – This track is anchored in a broad coalition of national institutions, CGIAR Centers, development partners, and private actors, each contributing complementary strengths to ensure delivery at scale. Internally, the Alliance CIAT (lead) will coordinate with ILRI, IFPRI, and other CGIAR centers active in Kenya, drawing on proven innovations in climate forecasting, resilient crops, soil fertility, livestock management, and digital delivery. These contributions are organized through aligned science programs—CA, SFP, SAAF, DTA and GESI—ensuring scientific rigor, inclusivity, and digital integration.

  At the national level, KALRO (advisory platform) and KMD (forecast provider), working with MOALD, provide institutional legitimacy and direct access to Kenya's 7M+ farmer registry. Counties are already requesting integration of KAOP/DCAS into their extension plans, with MOUs signed in places such as Homa Bay County, demonstrating demand.

  Strategic research partnerships reinforce the upstream and midstream pipeline. The Climate Hazards Center (UCSB) and FEWSNET, engaged through the CA, will improve KMD's forecasting skill and data assimilation. AIM4Scale will contribute AI-based subseasonal-to-seasonal (S2S) prediction and advisory assembly tools, helping generate multilingual, farmer-ready advisories at scale with human-in-the-loop validation. Engagement of Strathmore Agri-Food Innovation Centre will build national capacity for long-term sustainability.

The track leverages major bilateral investments: NAVCDP and FSRP (>$500M combined), AICCRA III (~$10M for Kenya 2026-30), and the FLLoCA program with county climate funds. IFAD's upcoming PPP for climate information services, co-financed by the Gates Foundation (~$2.75M), offers immediate alignment, building on IFAD's $250M+ portfolio in Kenya. Complementary initiatives like iSPARK (University of Leeds) provide research on sustainability and policy adaptation.

Downstream, private and NGO partners including Digital Green, Kuza, WFP-FtMA, PxD, and AGRA's VBAA network extend last-mile reach. Demand surveys show >80% of farmers report usefulness of SMS advisories, and 50% are willing to pay, confirming both farmer demand and partner incentives. Collectively, these partnerships ensure rapid scaling to six million farmers, institutionalization within Kenya's extension system, and financial sustainability beyond 2028.

- **Risks & Bottlenecks** –

Risk Category	**Description of Risk**	**Mitigation Strategy**
Financial	Over-reliance on donor financing; ~50% farmer willingness-to-pay limits revenue base.	Tiered pricing: free basic advisories + premium services; embed in MOALD/county budgets; private sponsorships (input firms, telco CSR); reduce costs to <US$1/farmer by 2028.
Political / Institutional	Devolved agriculture creates county fragmentation; electoral cycles disrupt continuity.	MOUs with counties embedded in NAVCDP/FSRP; endorsement in National Agricultural Extension Strategy; demonstrate early quick wins to build political champions.
Operational	Integration of diverse data streams; tailoring advisories to varied zones; risk of low uptake.	AIM4Scale's AI for S2S forecasts; SAFIC for capacity building; content boards to localize in 5+ languages; trusted extension agents; USSD polls and hotlines for feedback loops.
Social Inclusion	Exclusion of women, youth, pastoralists due to digital divide and literacy gaps.	PxD inclusive design: IVR in local dialects, community radio, group sessions; women-only trainings; youth "digital advisors"; gender-disaggregated monitoring.
Environmental	Risk of maladaptive practices from poorly designed advisories.	Expert validation of content; alignment with national CSA guidelines; human-in-the-loop AI review; compliance with Kenya's Data Protection Act for ethical data use.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

- **Readiness & Responsible Scaling** – The readiness to scale of the solution track is anchored in existing reach and infrastructure, whose integration creates a multiplier effect. KALRO already delivers agro-weather advisories to over one million farmers, while KMD generates forecasts across daily, 7-day, subseasonal, and seasonal intervals. Yet these assets are underutilized when siloed. With the national farmer registry of 7M+ farmers and county MOUs in place, the conditions now exist to connect these systems into an integrated advisory pipeline. Evidence from Kenya shows >80% of farmers find SMS advisories useful and around 50% are willing to pay modest fees, indicating demand and adoption readiness. Digital delivery models have proven far more cost-effective than traditional extension (<US$1 per farmer annually at scale). Responsible scaling is ensured through IVR in local dialects, gender- and youth-focused outreach, and pastoralist-specific modules. Environmental safeguards include expert validation of all advisories against national climate-smart guidelines, with AI outputs reviewed in a human-in-the-loop process.

- **Achievable target, Resources & Activities** – With S4I catalytic support and strong national partnerships, the track can achieve nationwide delivery of digital climate advisories to six million farmers by end-2028, covering all major agro-ecological zones. By 2027, three million farmers will be actively reached, with 50% women and 40% youth, and at least 2 million hectares managed under improved climate-smart practices. Advisories will span 20 crops, livestock thermal stress alerts, and rangeland decision-support, delivered via SMS, USSD, IVR, and 3,000+ trained agents.
  Resources required:
    - **Staff:** A multi-disciplinary team including data/ICT scientists, meteorologists, agronomists, livestock and gender experts. Youth agripreneurs will be engaged at county level and linked the SAFIC and other capacity building program.
    - **Institutional:** KALRO's KAOP/DCAS platform, KMD forecasts across daily–seasonal intervals, the farmer registry of 7M+, and county governments.
    - **Financial:** S4I support (~US$1.5M) for AI personalization, monitoring, and integration; leveraged investments from NAVCDP, FSRP, FLLoCA, and AICCRA III contributing >US$15M to training, outreach, and infrastructure support.
    - **Technical:** Forecast downscaling from AIM4Scale and UCSB/FEWS NET; analytics and QA from SAFIC; inclusive delivery design and A/B testing from PxD; secure data flows via Digital Green's FarmStack.
- **Key activities:**
    - Integrate KMD forecasts into KAOP with automated APIs and AI-enabled content assembly.
    - Expand decision-support modules to 20 crops plus livestock and rangeland services.
    - Train and activate 3,000 extension agents and agripreneurs across counties.
    - Launch push-and-pull services with multilingual IVR/SMS/USSD.
    - Conduct bi-annual HCD cycles with women, youth, and pastoralists.
    - Embed DCAS within NAVCDP/FSRP county workplans to secure sustained financing.

- **Timeline, Milestones & Monitoring** – Over three years, the track will (1) upgrade KAOP with forecast APIs and AI-enabled advisory generation, (2) expand advisory content to 20 crops, livestock, and rangeland systems, (3) scale delivery via SMS, USSD, IVR and 3,000+ agents, (4) institutionalize partnerships with counties, NAVCDP/FSRP, and IFAD's PPP model, and (5) embed monitoring and learning through PxD, and bi-annual reviews.

  2026 – Foundation and early rollout:

    - Integration of KMD forecasts and global models into KAOP via API, supported by Climate Action program(downscaling and forecast skill).

- Automated SMS/USSD/IVR advisories launched in 10 counties, with SFP and SAAF content embedded for crops and livestock.
- 1,000 extension agents and agripreneurs trained, with Gender Accelerator guidance to ensure inclusivity.
- Target: 1M farmers reached.

2027 – Expansion and institutional embedding:

- Decision-tree advisories expanded to 20 crops, livestock THI, and rangeland tools.
- Push-and-pull services rolled out nationally; voice/IVR in local languages.
- NAVCDP and FSRP county programs integrate DCAS; IFAD PPP model piloted for blended finance.
- Target: 3M farmers reached, 1.2M women/youth.

2028 – National scale and sustainability:

- Advisory coverage reaches 20+ counties; 3,000+ agents active with Digital Accelerator co-designing user tools.
- Institutionalization in National Agricultural Extension Strategy, with co-financing from government budgets and private partners.
- Target: 6M farmers reached, 70% receiving ≥10 advisories per season.

Monitoring & learning: A/B testing frameworks co-developed with PxD will evaluate message formats and uptake; Partner platform such as Digital Green's FarmStack will enable secure feedback integration. KAOP analytics, USSD polls, and hotlines will provide real-time monitoring, while annual outcome surveys capture adoption, yield, and equity impacts. Bi-annual joint learning workshops with CGIAR, KALRO, counties, IFAD, and WB programs will adapt strategies, ensuring continuous improvement and responsible scaling.

Optional Annexes

- IPSR Innovation Profile (if available)
- Scaling Readiness assessment results
- Scaling Scan results (if available)
- Prior project reports
- Media or other evidence of prior scaling work that can be accelerated
- MELIA studies, impact assessments, reports, publications
- Partner commitments (MOUs, LOIs, letters of support)

Annex 1: References/evidences/reports

- Beach, R.H., Milliken, F., Franzen, S., Lapidus, D. (2025). *Meta-analysis of digital information interventions on adoption, yields and income in Sub-Saharan Africa.* ScienceDirect. Link

- Fabregas, R., Kremer, M., Schilbach, F. (2019). *Realizing the potential of digital development.* Science, 366(6471), eaay3038. Link

- Basis (2025). *Willingness to Pay for Agro-Weather Messages Among Kenyan Farmers.* University of California, Davis. Link

- TomorrowNow (2024). *Weather Intelligence that Reaches the Last Mile: Impact Report from Kenya.* Link

- https://www.worldbank.org/en/results/2023/06/07/scaling-up-disruptive-technologies-for-agricultural-productivity-in-kenya

- Grossi, A., Ghosh, A. (2025) More than 1 million people reached with improved Participatory Scenario Planning processes in Kenya. Link

Annex 2: MoU/LOI

MEMORANDUM OF UNDERSTANDING
BETWEEN

KENYA AGRICULTURAL AND LIVESTOCK RESEARCH ORGANIZATION (KALRO)

AND

THE INTERNATIONALCENTER FOR TROPICAL AGRICULTURE (CIAT) on behalf of
THE ALLIANCE OF BIOVERSITY INTERNATIONAL AND CIAT (THE ALLIANCE)

FOR

Collaboration in Scientific Research, Knowledge Exchange, Capacity and Institutional Development

Staff Secondment Arrangement with KALRO

KENYA AGRICULTURAL & LIVESTOCK RESEARCH ORGANIZATION

When replying, please quote:
Our Ref: KALRO/ICT/STAFF/VOL.III **Date:** 31st January 2024

Country Representative
Alliance of Bioversity International and CIAT
ICIPE Duduville Campus, off Kasarani Road
P.O. Box 823 – 00621
Nairobi, Kenya.

RE: CONFIRMATION TO ENGAGE AND SECOND A DATA SCIENTIST TO KALRO

KALRO supports the secondment of qualified individuals for the Data Scientist position in Nairobi, Kenya, by the Alliance of Bioversity International and CIAT under the project agreement. This engagement is supported by the MoU between the two organizations, granting approval to advertise the position, conduct interviews, and select suitable candidate(s). The engagement will commence upon the full execution of the contract, with the Alliance of Bioversity International and CIAT responsible for remuneration in accordance with its terms of service.

Details of the Secondment:

- **Start Date:** The engagement will commence after all parties sign the contract.
- **Duration:** The contract will be issued for one year and may be renewed thereafter, subject to performance, funding availability, and programming plans and needs.
- **Scope of Work:** The duties and responsibilities as per the contract.
- **Compensation Agreement:** Alliance of Bioversity International and CIAT will compensate salary and benefits for this position with project funds. All benefits, including annual leave and sick pay, will remain as per the terms of employment.

The staff will report to and work at the KALRO headquarters office, including institutes and centers. They will submit monthly deliverable reports signed and approved by the Director, ICT KALRO.

All other terms and conditions in the contract agreement must align with Alliance Biodiversity and CIAT's scheme of service.

Yours faithfully

Salim Kinyimu
Director, ICT

KALRO HEADQUARTERS,
P.O. Box 57811-00200, Nairobi, KENYA. Tel: 254-020 4183301-20/ 254-020 4183720 Fax: 254-020 4183344
Website: www.kalro.org

KMD MoU under review (expected to be signed in November, 2025)

Alliance

MEMORANDUM OF UNDERSTANDING ("MOU")

This Memorandum of Understanding ("the MoU") is made as of this day of ____________ 2025, between **International Center for Tropical Agriculture** (herein referred to as "**CIAT")** on behalf of **the Alliance of Bioversity International and CIAT**, with its principal place of business located at Via di San Domenico, 1 - 00054 Rome, and hereinafter referred to as The **Alliance**, of the one part;

AND

Kenya Meteorological Department, inclusive of the **Institute for Meteorological Training and Research (IMTR)**, with its principal office at Dagoretti Corner, Ngong Road, Nairobi (herein referred to as "**KMD")**.

CIAT and KMD are jointly referred to as Parties and the term Party means either of them as the context so permits and the terms "Party" and "Parties" mean their respective and joint successor(s), and authorized representative(s) as the case may be.

WHEREAS

1. **CIAT** is a member of the CGIAR System and is a non-profit international organization constituted in Washington by an agreement dated 28 May 1986 between the World Bank and the United Nations Development Program (UNDP) and governed, in Colombia, by Law 29 of 1988.
2. The International Plant Genetic Resources Institute, which operates under the name **Bioversity International**, is an International Research Organization and member of the CGIAR System. Bioversity was established as an independent international organization by the 'Agreement on the Establishment of the International Plant Genetic Resources Institute' with the Italian Government in 1991. The agreement was revised in 2015.
3. **Bioversity International** and **CIAT** have formed a partnership known as the **Alliance of Bioversity International and CIAT** through a Partnership Agreement between the two international Centers, in force since January 1, 2020, having a single Director General and a common Board of Trustees as part of its governance structure. The Alliance delivers research-based solutions that harness agricultural biodiversity and sustainably transform food systems to improve people's lives. The Alliance's solutions address the global crises of malnutrition, climate change, biodiversity loss, and environmental degradation.
4. **KMD's** mandate is to provide timely and accurate weather and climate information, including early warnings, to ensure the safety of lives, protect property, and conserve the natural environment. This mandate is derived from Executive Orders on the structure and organization of the Kenyan government and the World Meteorological Organization (WMO) Convention. KMD's functions include, but are not limited to, data collection and monitoring, data processing and forecasting,